SQL Learning Journal #18 – SET OPERATIONS

1. UNION and UNION ALL

UNION

UNION combines the results of two or more SELECT queries into a single result table. If the same row appears in both tables, UNION returns it only once.

```
SELECT *
FROM product1_table

UNION

SELECT *
FROM product2_table;
```

product1_table

product_id	product	category_id
1	laptop	10
2	mouse	10
3	keyboard	10
4	coffee	15
5	tea	15
6	milk	15

product2_table

product_id	product	category_id
1	laptop	10
2	mouse	10
7	keyboard_a	15

Result

product_id	product	category_id
1	laptop	10
2	mouse	10
3	keyboard	10
4	coffee	15
5	tea	15
6	milk	15
7	keyboard_a	15

The duplicate rows (laptop and mouse) appear only once.

UNION ALL

UNION ALL also combines the results of two or more SELECT queries, but it keeps duplicate rows.

```
SELECT *
FROM product1_table

UNION ALL

SELECT *
```

```
FROM product2_table;
```

Result

product_id	product	category_id
1	laptop	10
2	mouse	10
3	keyboard	10
4	coffee	15
5	tea	15
6	milk	15
1	laptop	10
2	mouse	10
7	keyboard_a	15

The rows that appear in both tables are included twice.

Key Difference

- UNION → combines results and removes duplicate rows.
- UNION ALL → combines results and keeps duplicate rows.

2. INTERSECT

INTERSECT returns only the rows that exist in both result sets.

table_1

product_id	product	category_id
1	laptop	10
2	mouse	10
3	keyboard	10

table_2

product_id	product	category_id
1	laptop	10
4	coffee	15
5	tea	15
6	milk	15
7	keyboard_a	15

```
SELECT *
FROM table_1

INTERSECT

SELECT *
FROM table_2;
```

Result

product_id	product	category_id
1	laptop	10

The result contains only the row that is common to both tables.

3. EXCEPT

EXCEPT returns the rows that exist in the first result set but do not exist in the second result set.

table_1

product_id	product	category_id
1	laptop	10
2	mouse	10
3	keyboard	10

table_2

product_id	product	category_id
1	laptop	10
4	coffee	15
5	tea	15
6	milk	15
7	keyboard_a	15

```
SELECT *
FROM table_1

EXCEPT

SELECT *
FROM table_2;
```

Result

product_id	product	category_id
2	mouse	10
3	keyboard	10

The result contains the rows that are in table_1 but not in table_2.

Summary of SET OPERATIONS

Operation	Result
UNION	Combines two result sets and removes duplicates
UNION ALL	Combines two result sets and keeps duplicates
INTERSECT	Returns rows common to both result sets
EXCEPT	Returns rows in the first result set but not in the second

What I Learned Today

Today I learned how SQL set operations can be used to combine and compare groups of data. I learned that UNION combines two result sets while removing duplicate rows, whereas UNION ALL keeps duplicates. I also learned that INTERSECT returns the rows that are common to both result sets, while EXCEPT returns the rows that exist in the first result set but not in the second.

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